

Oxidation & Reduction

Poynting Vector $\vec{S}$. The rate at which the EM energy transfer per $1m^2 \rightarrow \vec{S} \Rightarrow \frac{J/s}{m^2} \Rightarrow W/m^2$ power density

$$E = 10 \hat{a}_t + 2$$

$$H = 0.2 \hat{a}_t - 2$$

$$\vec{S} = 10 \times 0.2 = \underline{2 W/m^2} (+y)$$

$$\vec{S} = \vec{E} \times \vec{H} \rightarrow \vec{S} = \vec{E} \times \frac{\vec{B}}{\mu} \Rightarrow \vec{S} = \frac{1}{\mu} \vec{E} \times \vec{B}$$

$$\frac{V}{m} \times \frac{A}{m} = \frac{VA}{m^2} = W/m^2$$

ex. 4

$$\vec{E} = 20\hat{i} + 10\hat{j} - 30\hat{k}, \vec{B} = 0.1\hat{i} + 0.4\hat{j} + 0.2\hat{k}$$

$$\vec{E} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 20 & 10 & -30 \\ 0.1 & 0.4 & 0.2 \end{vmatrix} = \hat{i}(2 + 12) - \hat{j}(4 + 3) + \hat{k}(8 - 1)$$

$$= 14\hat{i} - 7\hat{j} + 7\hat{k}$$

$$|\vec{E} \times \vec{B}| = \sqrt{14^2 + 7^2 + 7^2} = 17.146$$

$$S = \frac{|\vec{E} \times \vec{B}|}{\mu} = \frac{17.146}{4\pi \times 10^{-7}} = 1.36 \times 10^7 W/m^2$$

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$$E = 10 \text{ at } +z$$

$$H = 0.2 \text{ at } -z$$

$$\vec{S} = 10 \times 0.2 = \underline{2 \text{ W/m}^2} (+z)$$

$$\vec{S} = \vec{E} \times \vec{H} \rightarrow \vec{S} = \vec{E} \times \frac{\vec{B}}{\mu_0} \Rightarrow \vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

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Spherical area: $A = 4\pi r^2$

Circular area: $A = \pi r^2$

hemispherical area: $A = 2\pi r^2$

Circumference: $L_0 = 2\pi r$

Transmitter: T_x

Receiver: R_x



$H = \frac{B}{\mu} = \frac{E}{2\mu_0 c}$
 120π

$S_{av} \rightarrow I = \frac{\text{Power}}{\text{Area}}$
 Intensity

$I = \frac{E^2}{2\mu_0 c} = \frac{B^2 c}{2\mu_0}$
 240π

ex. 5

Airplane at 10 km receives a signal of intensity $10 \mu\text{W/m}^2$

① Find the amplitude of E : $\therefore I = \frac{E^2}{2\mu_0 c} \rightarrow E = \sqrt{I \cdot 2\mu_0 c} = \sqrt{10 \times 10^{-6} \times 240\pi} = 0.087 \text{ V/m}$

" " " " B : $B = \frac{E}{c} = \frac{0.087}{3 \times 10^8} = 2.9 \times 10^{-10} \text{ T}$

② if a transmitter sends a beam in hemisphere. Find the power

$$A = 2\pi r^2 = 2\pi \times (10000)^2 \\ = 628.32 \times 10^6 \text{ m}^2$$

$$I = \frac{P}{A} \rightarrow P = I \cdot A = 10 \times 10^{-6} \times 628.3 \times 10^6 = 6283.2 \text{ W}$$